Nat

1) Introduction / Construction — Inductive

$$\frac{}{\Gamma \vdash 0 : Nat}\ Nat_0\text{-}I \qquad \frac{\Gamma \vdash M : Nat}{\Gamma \vdash S(M) : Nat}\ Nat_S\text{-}I$$

```
kind Nat
data 0 :: Nat
data S :: Nat -> Nat
```

2) Elimination / Inductive destructure

$$\frac{\Gamma \vdash M : Nat \quad \Gamma \vdash P : A \quad \overbrace{\Gamma, x : A \vdash Q : A}^{\Gamma \vdash \lambda x.Q : A}}{\Gamma \vdash rec(P, Q)(M) : A}\ Nat\text{-}E$$

3) β rules $\quad (\Gamma \vdash \lambda x.Q : A)$

$$\frac{\Gamma \vdash P : A \quad \Gamma, x : A \vdash Q : A}{\Gamma \vdash rec(P, Q)(0) \equiv P : A}\ \beta\ Nat_0$$

kind declaration.

$$\frac{\Gamma \vdash M : Nat \quad \Gamma \vdash P : A \quad \Gamma, x : A \vdash Q : A]}{\Gamma \vdash rec(P, Q)(S(M)) \equiv [rec(P, Q)(M)/x]Q}\ \beta\ Nat_S$$

<u>example</u> let $M = S(S(S(0)))$

$$\underline{rec}(P,Q)(S(S(S(0))))$$

$$\equiv [\underline{rec}(P,Q)(S(S(0)))/x]\,Q$$

$$\equiv [\,[\underline{rec}(P,Q)(S(0))/x]\,Q/x\,]\,Q$$

$$\equiv [\,[\,[\underline{rec}(P,Q)(0)/x]\,Q/x\,]/x\,]\,Q$$

$$\equiv [\,[\,[P/x]\,Q/x\,]\,Q/x\,]\,Q$$

$$\equiv (\lambda x.Q)\big((\lambda x.Q)\,((\lambda x.Q)(P))\big)$$

$$\equiv \big((\lambda x.Q)\circ(\lambda x.Q)\circ(\lambda x.Q)\big)(P)$$

(β) $\underline{rec}(P,Q)(0) \equiv P$

$\underline{rec}(P,Q)(S(M)) \equiv [\underline{rec}(P,Q)(M)/x]\,Q$

```
class Rec (i :: Nat)
  instance p :: Rec 0
  instance q :: (Rec m) => Rec (S m)
```

η) $\Gamma, z : Nat \vdash M : A \quad (\simeq \Gamma \vdash \lambda z.M : Nat \supset A)$

$$\Gamma \vdash \overbrace{[0/x]M}^{(\lambda x.M)(0)} \equiv P : A$$

$$\Gamma, z : Nat \vdash \overbrace{[S(z)/x]M}^{(\lambda x.M)(S(z))} \equiv \overbrace{[M/x]Q}^{(\lambda x.Q)((\lambda x.M)(M))} \quad ?$$

η-Nat

$$\Gamma, z : Nat \vdash M \equiv \underline{rec}(P, Q)(z) : A$$

β) $\underline{rec}(P,Q)(0) \equiv P$

$\underline{rec}(P,Q)(S(M)) \equiv [\underline{rec}(P,Q)(M)/x]Q$

$$\left(\begin{array}{l} (\lambda x.\underline{rec}(P,Q))(0) \equiv P \\ \\ (\lambda x.\underline{rec}(P,Q))(S(M)) \\ \quad \equiv (\lambda x.Q)((\lambda x.\underline{rec}(P,Q))(M)) \end{array} \right)$$

<u>special case</u> (let $A = \mathrm{Nat}$)

$$z : \mathrm{Nat} \vdash \underline{\mathrm{rec}}(0, y.\, s(y))(z) \equiv z : \mathrm{Nat}$$

<u>commuting conversion</u>

$$\Gamma, z : \mathrm{Nat} \vdash [\underline{\mathrm{rec}}(z; 0, y.\, s(y)) / z]\, M$$
$$\equiv \underline{\mathrm{rec}}(z; [0/z]M, y.\, [s(y)/z]M)$$